

ScamPulse AI

Telecom Scam & Spam Pattern Intelligence

Industry: Telecommunications • call-protection & spam/scam monitoring context

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Batch Code • ANP - D7444

Industry reference: Truecaller (inspiration only; not an official project)

Data source: FTC Do Not Call consumer complaint records used in the project

Industry Overview

02

Selected industry: Telecommunications • focus on complaint intelligence for call-protection analysis

SELECTED INDUSTRY

Telecommunications

ScamPulse AI examines consumer-reported unwanted-call complaints as an analytical signal for telecom call-protection and spam/scam monitoring.

Project context

Time • Subject • Robocall • Geography
• Originating number

These signals are analyzed together rather than interpreted as isolated fields.

Why data analysis matters in this project

1

PATTERN DISCOVERY

Turn high-volume complaint rows into recurring patterns that are easier to monitor.

2

MONITORING

Compare dates, subjects, geography and robocall status to spot shifts in complaint activity.

3

PRIORITIZATION

Use recurrence and emerging-pattern signals to focus deeper review on the most relevant records.

PROJECT VIEW

The analysis layer converts complaint records into structured intelligence that can support monitoring, prioritization and future detection work.

Problem Statement

03

Why ScamPulse AI is needed

THE INDUSTRY PROBLEM

Large-scale consumer complaint records contain important patterns that are difficult to identify manually.

WHY IT MATTERS

Raw complaint records need structure before they can support consistent monitoring and analytical decisions.

WHO IS AFFECTED

Telecom users experience unwanted calls; analysts and call-protection workflows need usable pattern intelligence.

WHAT IS MISSING

A repeatable analysis layer that connects temporal, subject, robocall, geographic and originating-number signals.

INDUSTRY QUESTION How can a telecom call-protection platform convert complaint data into actionable pattern intelligence?

Interpretation rule: complaint frequency is a monitoring signal, not confirmation of fraud.

Project Objective

04

Five clear objectives for Sprint 1

- 1 Transform consumer-reported complaint records into structured analytical intelligence.
- 2 Identify recurring and emerging patterns across time, subject, robocall status, geography and originating number.
- 3 Measure complaint concentration and recurrence to support monitoring priorities.
- 4 **Build an interactive Plotly dashboard for fast exploration and decision support.**
- 5 Create a clean analytical foundation for future anomaly detection, risk scoring and machine-learning work.

Data Quality



Cleaning



EDA



Dashboard



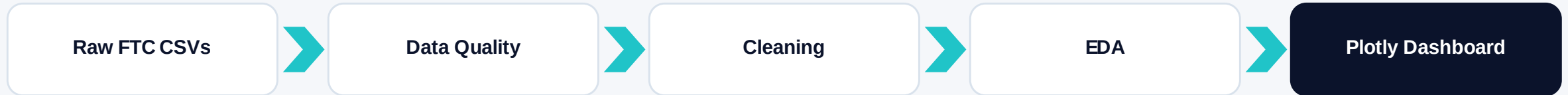
Future ML

Proposed Solution & Analysis Questions

05

Use Python to turn raw complaint records into a structured analytical workflow

PROPOSED SOLUTION



How Python helps

Pandas	clean, group, count, cross-tab and compare
NumPy	support numerical transformations and analysis
Matplotlib	validate EDA patterns with static charts
Plotly	present an interactive dashboard with hover/zoom

ANALYSIS QUESTIONS

- How does complaint volume vary by Created_Date?
- Which states, cities and area codes have the highest complaint volume?
- Which complaint subjects dominate the dataset?
- Which originating numbers recur most often?
- How much activity is marked as a robocall / recorded message?
- Which subjects, states and phone numbers are emerging in the recent period?

Dataset Information

FTC Do Not Call complaint records used for the project

DATASET NAME

FTC Do Not Call complaint records

Source Selected daily FTC complaint CSV files loaded by the project notebook

Coverage **31 Aug 2026 → 09 Sep 2026**

Format **CSV**

Selected files 8 daily CSV files: sep1, sep2, sep3, sep4, sep7, sep8, sep9, sep10

Important features

Company_Phone_Number • Created_Date • Violation_Date • Consumer_City • Consumer_State • Consumer_Area_Code • Subject • Recorded_Message_Or_Robocall

RAW DATASET

84,686 × 8

FINAL PROCESSED

84,678 × 10

UNIQUE SUBJECTS

15

Schema at a glance

- Company_Phone_Number
- Created_Date
- Violation_Date
- Consumer_City
- Consumer_State
- Consumer_Area_Code
- Subject
- Recorded_Message_Or_Robocall

Dataset description

Complaint records capture report timing, reported number, consumer geography, subject and robocall status.

Data Collection & Understanding

07

Notebook evidence from the supplied, executed Jupyter files

8 raw CSV files



Vertical concat



84,686 rows × 8

Jupyter Notebook — Data Understanding

```
data1.shape
```

```
{10399, 8}
```

```
data1.columns
```

```
Index(['Company_Phone_Number', 'Created_Date', 'Violation_Date',  
      'Consumer_City', 'Consumer_State', 'Consumer_Area_Code',  
      'Subject', 'Recorded_Message_Or_Robocall'],  
      dtype='object')
```

```
combined_data.shape
```

```
{84686, 8}
```

```
os.path.exists(<combined raw CSV path>)
```

```
True
```

Collection step: data1 ... data8 loaded from the raw folder and combined row-wise.

Jupyter Notebook — Data Quality

```
dataset.shape
```

```
{84686, 8}
```

```
dataset.isnull().sum()
```

```
Company_Phone_Number    3969  
Created_Date            0  
Violation_Date          0  
Consumer_City          27877  
Consumer_State          499  
Consumer_Area_Code      5  
Subject                 0  
Recorded_Message_Or_Robocall 4917
```

```
duplicate_count = dataset.duplicated().sum()
```

```
Number of duplicate rows: 0
```

```
dataset['Recorded_Message_Or_Robocall'].unique()
```

```
['Y' 'N' nan]
```

Quality checks observed: missing-value counts, exact duplicates and robocall values Y / N / missing.

Data Cleaning & Transformation

08

Actual cleaning decisions and final validation from the executed notebook

3,864 negative reporting delays retained for traceability and flagged through the date-quality field.

Jupyter Notebook — Data Cleaning & Validation

```
cleaned_dataset = cleaned_dataset.drop_duplicates()
print(cleaned_dataset.shape)
```

Shape after removing duplicates: (84678, 8)

```
print(cleaned_dataset.duplicated().sum())
```

0

```
print("Invalid phone numbers:", invalid_phone.sum())
```

Invalid phone numbers: 0

```
cleaned_dataset["Reporting_Delay_Days"] = (
    cleaned_dataset["Created_Date"] - cleaned_dataset["Violation_Date"]
).dt.days
```

Negative reporting delays: 3864

```
print("Final dataset shape:", cleaned_dataset.shape)
```

Final dataset shape: (84678, 10)

```
cleaned_dataset.isnull().sum()
```

Company_Phone_Number	0
Created_Date	0
Violation_Date	0
Consumer_City	0
Consumer_State	0
Consumer_Area_Code	0
Subject	0
Recorded_Message_Or_Robocall	0

ACTUAL STEPS PERFORMED

- 1 Remove duplicates**
8 exact duplicates removed; index reset.
- 2 Handle missing values**
Missing analytical fields represented as Not_Reported where appropriate.
- 3 Normalize identifiers**
Phone numbers validated as 10-digit strings; invalid/missing → Not_Reported.
- 4 Standardize categories**
City/state/text formatting standardized; robocall normalized to Y / N / Not_Reported.
- 5 Prepare dates**
Created_Date and Violation_Date converted to datetime; Invalid_Date_Order and Reporting_Delay_Days created.

FINAL DATASET
84,678 × 10

NULLS REMAINING
0

DUPLICATES
0

Exploratory Data Analysis (EDA)

09

Pandas + NumPy aggregations with Matplotlib outputs from the notebook

Temporal

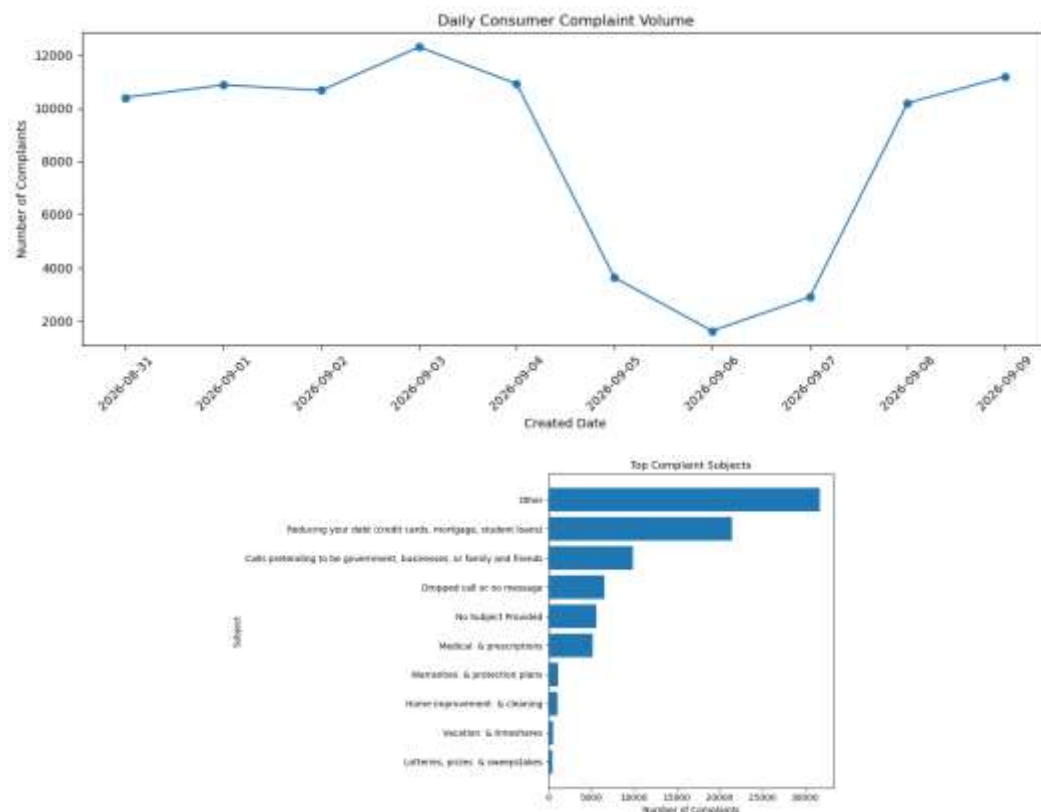
Subjects

Robocall

Geography

Numbers

Emerging



Matplotlib outputs: daily complaint volume + top complaint subjects

Important findings

PEAK DAY

03 Sep 2026 • 12,296

LOWEST DAY

06 Sep 2026 • 1,628

LEADING SUBJECT

Other • 31,724 • 37.46%

DEBT-RELATED SUBJECT

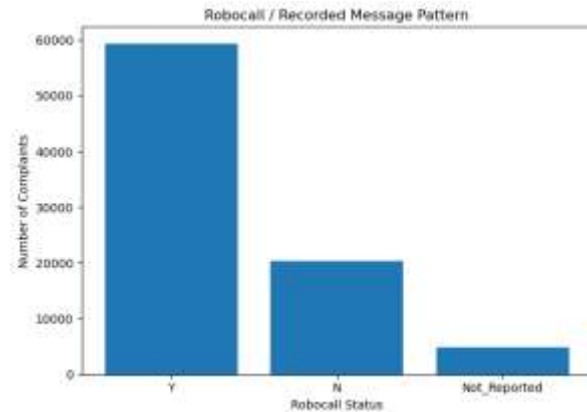
21,465 • 25.35%

The EDA also covered time-of-day, day-of-week, subject × robocall, state/city/area-code concentration, originating-number recurrence, reporting delay and emerging-pattern comparisons.

Data Visualization

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Actual chart outputs from the project notebook — Chart → Purpose → What it shows → Key insight

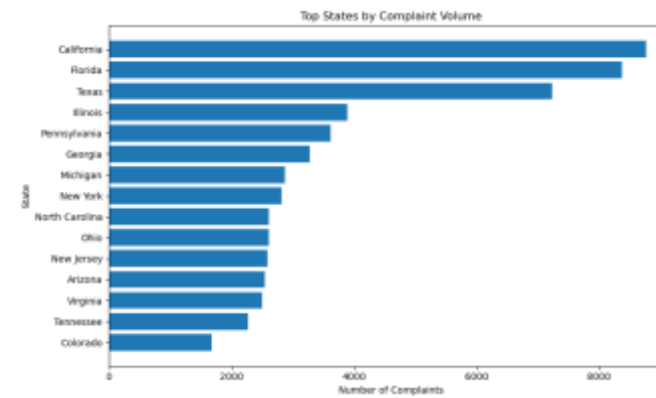


Robocall / Recorded Message Pattern

Purpose: compare Y / N / Not_Reported status.

Shows: 59,405 Y • 20,357 N • 4,916 Not_Reported.

Key insight: 70.2% of complaint records are marked Y.

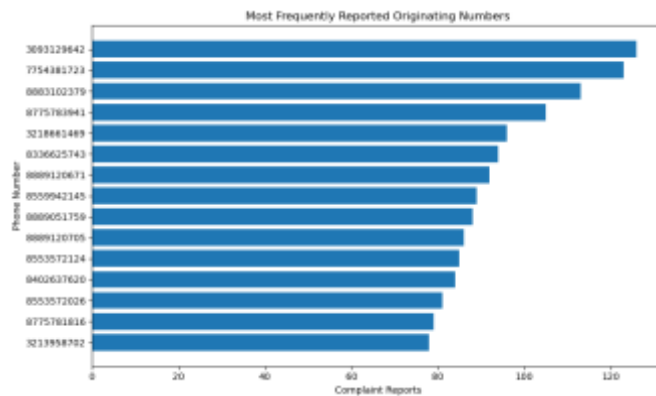


Top States by Complaint Volume

Purpose: identify geographic concentration.

Shows: top states ranked by complaint volume.

Key insight: California is the top state in the analyzed records.

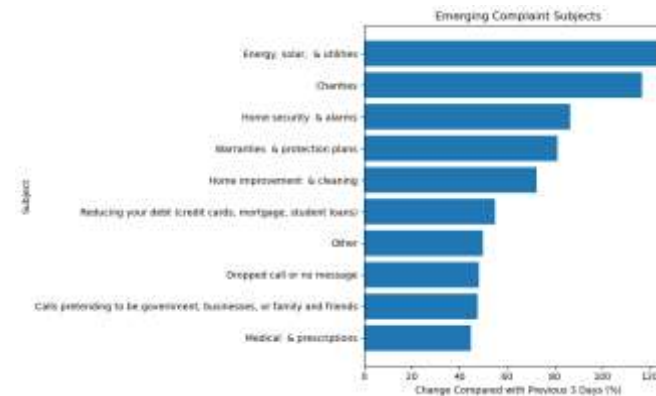


Most Frequently Reported Originating Numbers

Purpose: detect recurrence among reported numbers.

Shows: the most frequently reported originating numbers.

Key insight: 70,265 distinct reported numbers; 4,767 were reported more than once.



Emerging Complaint Subjects

Purpose: compare recent 3 available dates vs previous 3.

Shows: subjects with increasing recent-period counts.

Key insight: "Other" increased by +3,053 reports.

Static charts are exact notebook outputs; dashboard adds Plotly interactivity for the same analysis.

Key Insights & Recommendations

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What the actual dataset analysis supports

KEY INSIGHTS

01 Complaint activity varies sharply by date: 12,296 on 03 Sep vs 1,628 on 06 Sep.

02 "Other" is the largest subject: 31,724 reports (37.46%).

03 Debt-related complaints contribute 21,465 reports (25.35%).

04 70.2% of records are marked as recorded-message / robocall (Y).

05 California is the top state; 4,767 originating numbers recur in the dataset.

RECOMMENDATIONS

01 Monitor daily volume with a trend/rolling-average view to surface spikes and drops.

02 Prioritize high-volume subjects for deeper pattern review.

03 Use robocall status as one monitoring dimension alongside subject and time.

04 Prioritize repeated and emerging numbers for review — do not label them as confirmed fraud from complaint frequency alone.

05 Retain date-quality flags and use valid reporting delays for delay-specific analysis.

Interpretation rule: consumer complaint frequency is a monitoring signal, not confirmation of fraud.

Conclusion

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Sprint 1 turns raw complaint data into a reusable analytical foundation

WHAT WAS ACHIEVED

A reproducible path from raw complaint records to cleaned data, deep EDA and an interactive Plotly dashboard.

Raw complaint records

Cleaned dataset

EDA

Interactive dashboard

Major findings

84,678 processed rows • 70.2% robocall-marked • 4,767 repeated numbers

FUTURE SCOPE

From analytics to intelligent detection

1

Feature engineering

2

Anomaly detection

3

Risk scoring

4

ML classifier

5

Deployment

BOTTOM LINE

The Sprint 1 output creates a structured base for future anomaly detection, risk scoring and machine-learning-based spam/ scam monitoring.

THANK YOU

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